

Reimagining Airbnb

for the Modern Traveller & Community

High Level System Design: Exploring New Possibilities

Product Manager Assignment · Airbnb

What We're Solving

Two key problems · One connected platform

01

Hassle-Free Check-In

Digital identity verification via DigiLocker API or secure ID upload — so guests check in safely without friction.

02

Intergenerational Housing

A platform that connects elderly homeowners with individuals seeking affordable housing in exchange for daily assistance.

The Problem

Where today's Airbnb experience falls short



Check-In Friction

Guests face unsafe and inefficient ID verification processes during check-in. There's no standardised digital identity layer, leaving both hosts and guests vulnerable.



Elderly Housing Gap

Millions of elderly homeowners have spare rooms and need daily assistance, while families and individuals struggle to find affordable housing — yet no platform bridges this gap meaningfully.



Missing Intergenerational Layer

Airbnb's current model is transactional and short-term. There's no infrastructure for community-driven, long-term living arrangements that create mutual value.

Hassle-Free Check-In

Option A — Recommended

DigiLocker API Integration

- Guest grants Airbnb one-time access via DigiLocker
- OTP-based authentication — user consent at every step
- Airbnb never stores raw ID; only a verification token is issued
- Linked directly to the booking ID for traceability
- Fully compliant with data privacy standards

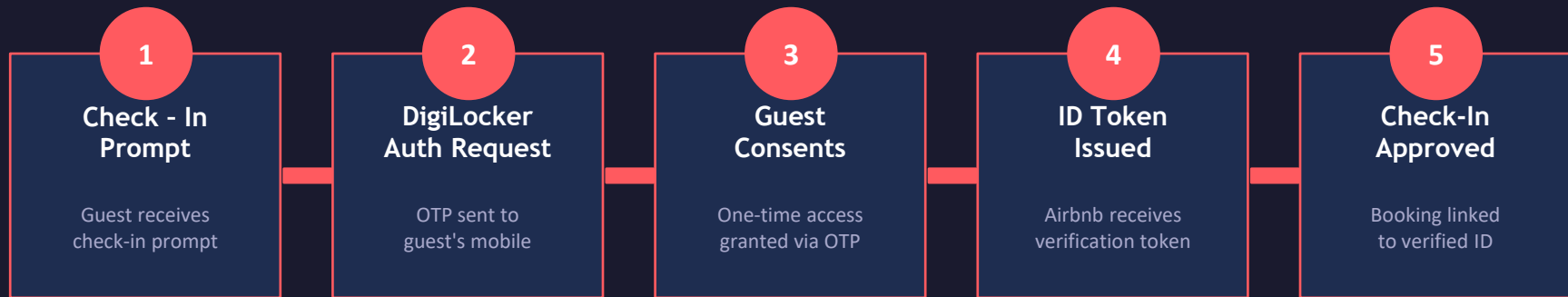
Option B — Fallback

Secure Manual ID Upload

- Guest uploads a digital copy of their ID within the app
- File is encrypted end-to-end and stored temporarily
- Automatically deleted 48 hours post check-in
- Hosts receive a verified badge — not the raw document
- Available for users without DigiLocker access

How the Check-In Flow Works

DigiLocker API — Step by step



No raw ID stored by Airbnb

User consent at every step

OTP-authenticated access

Token linked to booking ID

Intergenerational Housing Platform

 *The Concept: Elderly homeowners list spare rooms at low/subsidised rent. In return, tenants agree to help with defined daily tasks — cooking, grocery runs, or errands — creating mutual value and reducing isolation on both sides.*

Elderly Hosts

- Create a profile listing their spare room
- Specify type of help needed (e.g. cooking)
- Set a small rental fee for the arrangement
- Review tenant applications & background checks

Tenant Seekers

- Browse listings based on location & fit
- Apply with profile + background consent form
- Agree on a written task commitment list
- Rate the experience post-stay

Platform Safety

- Background checks before any approval
- Digital task agreement signed in-app
- Weekly check-in nudges for both parties
- Emergency contacts and escalation path

High-Level System Architecture

Four interconnected modules powering the platform

User Interface Layer

- Profile creation & management
- Accommodation search & apply
- In-app messaging & notifications
- Review & rating system

Background Check System

- Identity verification (DigiLocker/upload)
- Reference & history checks for tenants
- Results surfaced to hosts — not raw IDs
- Ongoing compliance monitoring

Listing & Management

- Elderly host profiles & room listings
- Tenant application tracking
- Booking & selection workflow
- Calendar & occupancy management

Assistance Management

- Digital task agreement templates
- Chore completion tracking
- Automated check-in reminders
- Dispute resolution support

Data Flow Within the System

How information moves across modules to serve users seamlessly

01

Host Listing Flow

Elderly host creates a profile → Listing published to search → Tenant applies → Host reviews application via UI

02

Background Check Flow

Tenant grants DigiLocker/upload consent → Background check system verifies identity → Verification result surfaced to host dashboard

03

Tenant Selection Flow

Host reviews verified profiles → Selects best-fit tenant → Accommodation system updates booking status → Both parties notified

04

Assistance Agreement Flow

Both parties agree to task list digitally → Assistance system tracks weekly task completion → Ratings submitted on conclusion of stay

Benefits & User Impact

Who benefits — and how

Elderly Hosts

- Affordable companionship model
- Daily help with household tasks
- Extra income from low rental fee
- Safety through background-checked tenants

Tenant Seekers

- Access to affordable housing options
- Flexible, community-style living
- Skill exchange & human connection
- No hefty security deposits

Platform Trust

- Verified identities build host confidence
- DigiLocker means no raw data stored
- Transparent task agreements reduce disputes
- Ratings create accountability

Society at Large

- Reduces elderly isolation & loneliness
- Makes cities more inclusive & affordable
- Builds intergenerational social bonds
- Sustainable housing model at scale

Conclusion

A Smarter, Safer, More Human Airbnb

The proposed system upgrades Airbnb across two critical dimensions: a safer, consent-driven check-in experience powered by DigiLocker and secure ID upload; and a first-of-its-kind intergenerational housing platform that turns spare rooms into community bonds.

DigiLocker API

OTP Authentication

Background Checks

Task Tracking

Community Trust

Meet Chauhan · Product Manager · Airbnb

Note: Based on hypothetical situations and assumptions